SEMICONDUCTOR / PLASMA / PROCESS MONITORING

# Evaluate the alarm. Not only the classifier.

Optical emission spectroscopy, model-conditional tomography and controlled monitoring stress tests—separating synthetic evidence from measured-data calibration.

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**50% to 13.56%**

Mean false-positive rate
Transferred recalibrated

**61.09% to 99.39%**

Mean unconditional recall
Same future evaluation frames

**37,814**

Distinct future frames
3 sessions / 6 reused roles

Public measured N2 study; 512 dataset-labelled normal reference frames. See pp. 6-7.

Research-to-engineering focus. I applied an established laboratory CT-OES framework, evaluated its assumptions, and developed a portfolio extension to test synthetic state changes and alarm robustness. A separate public measured N2 study shows why a near-perfect AUC need not yield reliable transferred thresholds.

**SKILLS DEMONSTRATED / MEASUREMENT**

OES · Gaussian fitting · inverse problems · radiation-trapping assumptions · Python analysis

**SKILLS DEMONSTRATED / EVALUATION**

Parent-grouped synthetic comparisons · PCA/SVM · correlated-noise tests · threshold transfer · abstention cost

Not a production-fab qualification or real-fault detector. No actual yield uplift, fab process control, or Ar-to-$N_2$ transfer has been demonstrated. Data provenance is kept explicit on each page.

01 / OPTICAL EMISSION DIAGNOSTICS

# From measured spectra to spatial maps

The starting point was real optical-emission analysis—not a synthetic semiconductor fault dataset.

| | |
|---|---|
| **312 viewing channels**<br>Measured Ar spectra | **2,808 line fits**<br>9 emission lines |
| **CT reconstruction**<br>31 × 31 grid | **Plasma interpretation**<br>Model-conditional |

**118**
quality-review flags
Not predictive failures

**31 × 31**
spatial reconstruction grid

**50°**
code-defined LoS span
−25° to +25°

## My CT-OES contribution at SNU

I applied the laboratory's existing Python CT-OES framework, organised the channel inputs and exclusions, examined Gaussian fitting and quality flags, and interpreted regularised reconstructions. I compared relative spatial trends with supplied Langmuir-probe measurements while tracking the differences in resolution, perturbation and calibration assumptions.

**SIGNAL PROCESSING**

Constrained multi-line Gaussian fitting · overlap handling · fit review · background and calibration assumptions.

**INVERSE RECONSTRUCTION**

Ray geometry · nonnegative regularisation · projection-versus-field error · edge artefacts.

## Why the quality-control count matters

Of 2,808 spectral fitting tasks, 118 received quality-review flags. That is an **analysis workload and QC metric**, not a model accuracy score or a claim that 118 devices would fail.

> **Attribution boundary.** The CT framework predated my placement; applying it, investigating reconstruction results and creating the follow-on portfolio evaluations are my contributions. I do not claim to have invented the laboratory reconstruction platform.

[S1] SNU research summary and PlasmaSense run_summary.json (v0.10.0); geometry is a code configuration, not independent metrology of the original apparatus. Original chamber raw data are not reproduced here.

02 / RECONSTRUCTION VALIDATION

# The map is not the measurement.

An inverse solution must be evaluated in the reconstructed field, with model assumptions and rejected regions visible.

## Reconstruction accuracy cannot be inferred from a small projection residual

**29.23%**
field relative L2 error
Noiseless same-forward-model phantom

**2.176 eV**
model-conditional $T_e$ median
412 accepted pixels

**0.45 m**
assumed effective path
Not an independently measured path

The noiseless phantom check showed that even when a solver reproduced line-of-sight projections, the recovered **spatial field could still be substantially wrong**. Limited-angle geometry and regularisation make field fidelity a separate validation question.

## Temperature and pixel acceptance

| Pixel status | Count | What it means |
|---|---|---|
| Accepted model-conditional $T_e$ pixels | 412 | Used for the reported 2.176 eV median |
| Metastable fits rejected or at bounds | 303 | Not presented as reliable temperature estimates |
| Outside or invalid measurement support | 245 | Not assigned reconstructed physical conditions |
| Temperature root not bracketed | 1 | No accepted root under chosen assumptions |
| **Total grid** | **961** | 31 × 31 pixels |

> **Do not conflate historical and new results.** An earlier Donnelly/Dasgupta-based OES analysis reported a 1.42 eV representative value; the later recovered-BSR dataset, path and spectral-response assumptions yield 2.176 eV in PlasmaSense. They are **different model-conditioned estimates**, not two verified observations of the same absolute physical quantity.

[S1] run_summary.json: atomic_dataset, geometry, diagnostics and phantom_relative_l2_error; results/tables/plasma_diagnostics.csv. No independently validated absolute electron density is claimed.

03 / PHYSICS-INFORMED EVALUATION

# Controlled states. Not invented fab faults.

The simulator tests a deliberately limited scientific question and keeps synthetic labels separate from observed operating outcomes.

## What PlasmaSense actually simulates

The extension applies **reduced-physics interventions** to an experimental Ar emission template. Changes to effective electron energy and relative electron density produce labelled synthetic responses under declared excitation and observation assumptions.

**Measured anchor**
One Ar condition

**Imposed states**
Energy · density · local variation

**Synthetic signals**
Shared geometry and template

**Detector comparison**
Held-out parent groups

## Four detector approaches

| Comparator | Feature or training strategy | Purpose |
|---|---|---|
| Original PCA | Original projection/signal representation | Baseline unsupervised anomaly response |
| Mean-feature PCA | Mean signal features for each line | Test reduced feature sensitivity |
| Noise-aware mean PCA | Normal-noise augmentation during development | Test robustness to independent channel noise |
| Physics-trained SVM | Supervised examples from imposed model states | Test controlled-state discrimination |

**SPLIT DISCIPLINE**

Synthetic children of a common parent remain in one partition; learning and normal-reference scores are fixed before changed-noise evaluation.

**CRITICAL LIMITS**

The generator is not driven by real RF power, pressure or gas recipe. Uniform relative density scaling is confounded with optical gain.

**What the study answers.** It compares recognition of stated synthetic changes and stress sensitivity. It does **not** demonstrate real fab fault detection, optical hardware robustness or universally superior physics-trained learning.

[S2] Saved grouped evaluations. Demo inventory: 2,520 replay cases / 10,080 detector decisions; these are interface verification counts, not independent experiments.

04 / COVARIANCE STRESS

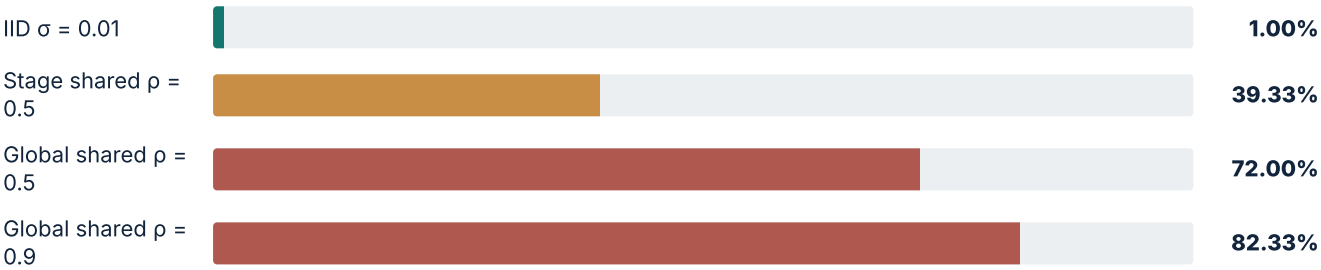
# The failure case matters most.

A lower false-alarm rate under independent noise was not a general guarantee of reliable monitoring.

## Noise robustness: where a promising method failed

Under the stated Maxwellian-response synthetic regime, normal-noise augmentation reduced IID false alarms in one stress setting. But correlated noise—plausible when measurement channels share optical or acquisition disturbances—exposed a major failure mode.

MEAN FPR · NOISE-AWARE MEAN-FEATURE PCA




For a separate IID high-noise condition, original PCA FPR was 100% and noise-aware mean PCA was 8% (mild-change recall 62.22%). The improved IID result did not transfer to these correlated-noise shifts. [S2]

| Condition | Observed comparison | Important trade-off |
|---|---|---|
| IID σ 0.01 | Original PCA FPR 2.67%; noise-aware 1.00% | Mild-change recall: 62.56% → 57.78% |
| IID σ 0.03 | Original PCA FPR 100%; noise-aware 8.00% | Noise-aware mild recall 62.22% |
| Stage-shared ρ 0.5; σ 0.01 | Noise-aware FPR 39.33% | Calibrated IID robustness breaks under covariance change |
| Global-shared ρ 0.5 / 0.9 | Noise-aware FPR 72% / 82.33% | Shared-channel effects overwhelm the normal reference |

**Decision.** Do not call a model “noise robust” merely because IID tests improve. Retest the correlation structure of normal measurement errors and report both false alarms and the sensitivity lost on subtle state changes.

[S2] state_improvement_families and robustness comparison outputs; means over five simulator seeds around one experimental anchor. Not confidence intervals, independent physical experiments or measured industrial fault rates.

05 / MEASURED-DATA ALARM TRANSFER

# High AUC. Poor transferred threshold.

A public measured dataset tests calibration transfer on unchanged future evaluation populations.

**Separate measured evidence.** To test threshold transfer, I used a public $N_2$ controlled-injection OES dataset containing **52,393 spectra from three physical sessions**. This is not the Ar state generator and does not validate Ar-to-$N_2$ fault transfer.

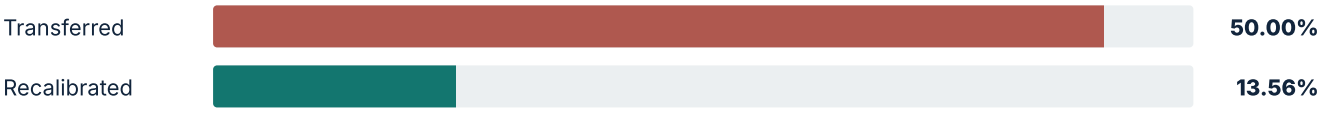
**0.9989**
global mean AUC for author-index LR
Six reused role permutations

**37,814**
distinct matched future frames
Calibration evaluation

**512**
dataset-labelled normal frames
Reference acquired before evaluated frames

## Same future frames, different threshold policy

MEAN FALSE POSITIVE RATE




MEAN LEAK RECALL




**Interpretation.** A nearly perfect score ranking did not define a transferable decision threshold. A target-session labelled-normal reference improved alarm operating points, but one session still produced 33.22% false alarms.

**Critical denominator:** three physical sessions were reused in six train/test role permutations; averages are descriptive. The broader AUC is 0.998915, while the matched calibration-subset AUC is 0.998360. Do not quote both as one exact evaluation. Acquisition frames and early leaks before the 512-frame reference were excluded. [S3]

06 / DECISION POLICY

# Do not optimise away the events that matter.

The best-looking false-positive metric is not automatically the safest or most operationally useful policy.

## A cleaner alarm chart can hide a dangerous loss

| Policy | Mean FPR | Mean unconditional recall | Leak frames withheld |
|---|---|---|---|
| Recalibrated, no guard | 13.56% | 99.39% | 0% |
| Add 3% support guard | 9.97% | 52.56% | 46.83% |

MEAN UNCONDITIONAL LEAK RECALL




**AT FIRST GLANCE**

False alarms fall by 3.59 percentage points: 13.56% → 9.97%.

**THE HIDDEN COST**

Nearly half of leak frames are withheld; unconditional recall falls by about 46.83 percentage points.

**Engineering decision: no automatic veto.** I retained the guard as a **review/uncertainty signal**, rather than using it to suppress a primary alarm. A high accepted-only recall does not account for real events deferred by the system.

## Limitations that remain

Valve-operation and temporal effects were not causally isolated from leak labels. Channel 127 cannot be assigned an unambiguous wavelength due to axis/column alignment differences. The 512-frame normal reference is **dataset-labelled**, not externally proven by independent fab sensors. Threshold choices were explored during development on inspected data; no newly held-out factory validation is claimed.

[S3] external_calibration/summary.csv; external_guard/summary.csv and metrics.csv. All FPR/recall/guard figures above are means across reused role permutations. Recall is unconditional and counts withheld leak frames as not alarmed.

07 / OWNERSHIP AND RELEVANCE

# A traceable contribution—from data to decision.

Clear ownership and negative results are part of the technical evidence, not caveats hidden at the end.

## What I contributed

| Phase | My documented work |
|---|---|
| Diagnostic foundation | Applied the SNU CT-OES framework; reviewed channel ordering, Gaussian fitting and spectral QC. |
| Inverse-model checks | Kept geometry explicit, investigated regularised field errors and temperature assumption sensitivity. |
| Physics-informed evaluation | Defined reduced-physics interventions, parent-grouped splits and four detector comparisons. |
| Noise robustness | Evaluated fixed detector/calibration settings under IID, stage-shared and global-shared channel noise. |
| Measured benchmark | Tested threshold transfer and normal-reference calibration on public $N_2$ sessions and matched future frames. |
| Decision and communication | Measured guard abstention cost; documented reasons to keep review flags rather than auto-suppress alarms. |

**PROJECT OWNERSHIP**

PlasmaSense is an independent portfolio extension from supervised research. I developed the analysis framing, evaluated the results and documented the engineering decisions.

**WHAT IS NOT CLAIMED**

A new CT framework at SNU, independently operated ALD/XPS tools, real fab fault data, validated yield improvement or a production digital twin.

## How this relates to semiconductor process engineering

These results motivate a practical workflow: inspect spectrometer/optics QC for signal failures; obtain a labelled or operationally confirmed reference when transfer shifts; and investigate persistent alerts with tool logs and wafer metrology **before** changing a recipe. These are recommended checks, not deployed fab procedures.

PLASMA DIAGNOSTICS · PYTHON · SPECTRAL FITTING · CT-OES · PCA / SVM · NOISE ROBUSTNESS · MODEL VALIDATION

08 / ROLE ALIGNMENT & SOURCES

# Ready for a technical conversation.

A concise explanation of scope, limitations and decision-making is as important as the underlying numbers.

## Skills demonstrated

PROCESS DIAGNOSTICS

My work connects plasma spectroscopy with monitoring reliability. I applied CT-OES analysis to 312 measurement channels, then tested how model assumptions, channel noise and threshold transfer affect the trustworthiness of a monitoring decision. I would bring that diagnostic discipline to process and yield investigations.

MODEL EVALUATION

My background combines low-temperature plasma diagnostics, numerical reconstruction and careful experimental interpretation. In my PlasmaSense extension I tested controlled synthetic changes and identified correlated-noise failure conditions; this motivates independent verification before drawing conclusions about manufacturing process health.

ADVANCED MANUFACTURING / EQUIPMENT

I enjoy translating difficult measurement and modelling results into practical engineering checks. My projects cover quantitative spectral fitting, limited-angle reconstruction and measured-data alarm calibration, with explicit attention to false positives, missed events and model boundaries.

Methods and source roles
Schulze et al. (2008), J. Phys. D 41, 065206. Common-upper ratios and escape factors. DOI 10.1088/0022-3727/41/6/065206.
Chouteau et al. (2024), PSST 33, 075016. Eq. (8) metastable balance; Eqs. (10)-(11) relative ne. DOI 10.1088/1361-6595/ad5d11.
Donnelly (2004), J. Phys. D 37, R217. Historical Fig. 5 thick direct-plus-cascade input. DOI 10.1088/0022-3727/37/19/R01.
Dasgupta et al. (published 1999), Phys. Rev. A 61, 012703. Fig. 6 theoretical 1s5-to-2p channels. DOI 10.1103/PhysRevA.61.012703.
Boffard et al. (2012), J. Phys. D 45, 045201. Osaka 420.1/419.8 nm method only. DOI 10.1088/0022-3727/45/4/045201.
Public N2 data: Youngjun Yuk, Mendeley Data v1, DOI 10.17632/9gy9v3h4mp.1, CC BY 4.0. Separate retrospective label-assisted evaluation.
Donnelly primary lineage: Schappe et al. (1994), PRA 50, 444; Chutjian & Cartwright (1981), PRA 23, 2178. These historical method references are distinct from the later demo's model inputs and do not certify its numerical results.

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**Version 1.2 / 31 October 2026.** Graduate research portfolio. Charts reproduce reported results; no new experimental or operational performance certification.

# Read the methods. Trace the decision.

### Measured diagnostics vs synthetic replay

The diagnostic workload contains 9 lines x 312 channels. The saved detector replay uses 8 line features: 696.54, 706.72, 738.40, 750.38, 763.51, 800.62, 801.48 and 842.47 nm. These are different analysis scopes; the replay does not display all diagnostic fits.

### Frozen / mean-feature PCA

Frozen means fitted models and normal calibration scores remain fixed during noise stress. Mean-feature PCA uses eight line-wise means of log(signal/reference) across 312 channels.

### Noise correlation and response assumptions

Rho denotes imposed same-line channel log-noise correlation: within a 78-channel stage or across all 312 channels. Maxwell refers to a Maxwellian electron-energy response. EEDF shape variation is a sensitivity study; its energy parameter is not automatically Te.

### Same model / different operating policy

The measured benchmark uses an author-index single-channel logistic classifier. Target-session normal-reference calibration changes its threshold; the support guard withholds decisions. The displayed N2 policy comparison does not retrain a model.

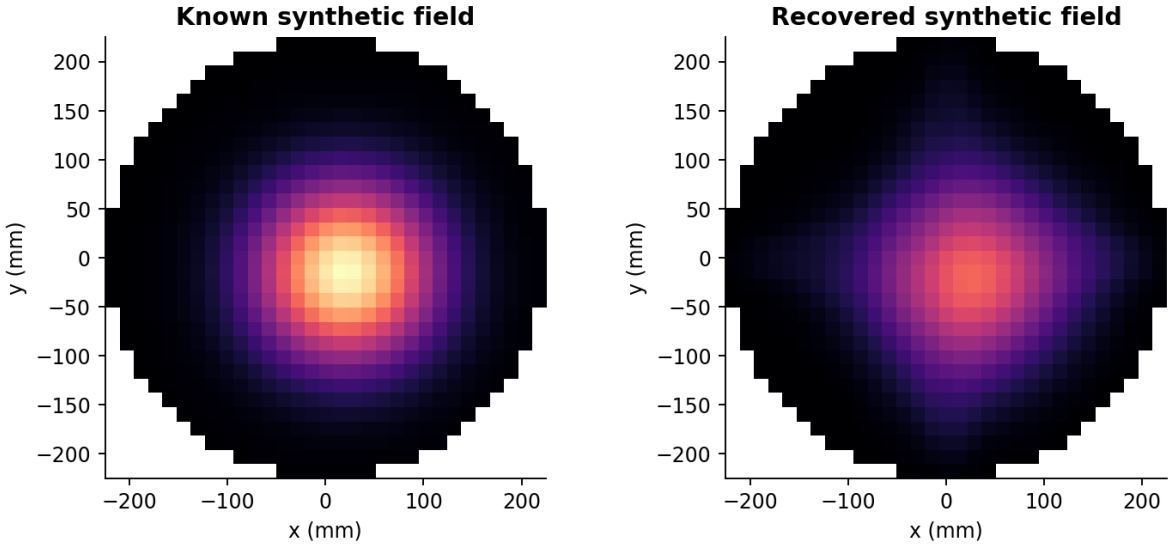
### Reproducibility and evaluation scope

The reported replay comprises 2,520 saved cases and 10,080 detector decisions. These are consistency-check counts, not independent experiments or held-out scientific validation. This application edition presents results in the portfolio; no interactive lab or downloadable research data is included.

# Evidence behind the decision

Selected results / technical review

Phantom validation · same forward model · synthetic, not an experiment




**29.23% field relative L2 error** in a noiseless, same-forward-model phantom. This synthetic check reveals smoothing and field distortion; a small projection residual does not establish spatial accuracy.

## NOISE STRUCTURE CHANGES THE OUTCOME

At σ = 0.01, noise-aware mean PCA FPR rose from 1% under independent noise to 39.33% with stage-shared correlation ρ = 0.5 and 72% with global-shared ρ = 0.5. Five reused seeds, assumed covariance and fixed models/thresholds. Physics-trained SVM was not uniformly superior.

SCOPE OF THIS APPLICATION EDITION

This portfolio presents previously reported aggregate results and selected figures. The compact website includes the Highlights and technical PDF; research CSVs, source code and the interactive lab are not included.

**Boundaries:** $N_2$ valve/time effects remain unresolved. Ar synthetic and $N_2$ measured studies are distinct. Absolute $n_e$ and pressure/power/gas-ratio prediction are unsupported. Laboratory CT engine: existing framework; SNU analysis and portfolio evaluation: my work.

Sources: retained PlasmaSense evaluation results; Youngjun Yuk, Mendeley Data v1, DOI 10.17632/9gy9v3h4mp.1 (CC BY 4.0). Public dataset attribution does not imply that private laboratory data are distributed.